

The Research on Image Classification Based on Convolutional Neural Network

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ABSTRACT

Convolutional neural network (CNN) performs well in image classification and segmentation, target detection and other applications. Researchers are paying more and more attention to its powerful feature learning and feature expression abilities. Based on CIFAR10 data set, this paper used CNN for image recognition. In the experiment, the training accuracy and testing accuracy of CNN can be improved by increasing the training times and the number of full connection layers of CNN. In this paper, if training 1000 epochs, and use 2 full connections layers. the training accuracy can reach 91%, and the testing accuracy can reach 90.5%.

CCS CONCEPTS

• Computing methodologies → Scene understanding.

KEYWORDS

Image Classification, Convolutional Neural Network, CARAR-10

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1 INTRODUCTION

As an important branch of computer vision, image recognition is the beginning of research. It was acquired in the 1940s and in the 1960s with the advent of artificial intelligence rapid development. In the past few decades, the application field of image recognition has been covered military, security, biomedicine, agriculture, automation, and many other human social lives. Convolutional Neural Network (CNN) is one of the best algorithms to accomplish the task of image

recognition. The Convolutional Neural Network is designed to imitate the learning mode of human beings. Through the training and testing of the input samples, the characteristics are extracted from simple to deep to distinguish the samples. Neural network can reduce the error of image classification and get high recognition rate.

This project is based on convolutional neural network for image recognition. The accuracy of image classification can be improved by changing the network model structure, parameter meaning and network hierarchy. The model is applied to image classification.

2 RELATED WORK

In paper [3], authors proposed a modified VGG-16 network and used that model classify CIFAR-10. The proposed model achieved 8.45% error rate on CIFAR-10, and it without severe overfitting. The paper [1] classify image using CNNs on the embedded systems. the embedded systems only a limited amount of memory is available. The results shows that 85.9% image classification accuracy is obtained by their framework while requiring 2GB memory only. The paper [2] proposed a recurrent convolutional neural network (RCNN) for (static) object recognition. The paper [4] proposed a method to dynamically set the pruning rate according to the situation of each layer. On CIFAR-10, the network achieved a compression ratio of 52.70% on ResNET-56 and only a 0.13% increase on top-1 errors. In paper [5] state-of-the-art convolutional neural networks viz. DenseNet, VGG, Residual Network and Inception (v3) Network are compared on CIFAR-10. The accuracy of 82.68%, 88.79%, 81.01%, and 84.92% respectively.

3 CONVOLUTIONAL NEURAL NETWORK

CNN has achieved great success in many research fields, such as speech recognition, image recognition, image segmentation, natural language processing and so on. Although the problems solved in these fields are different, all these application methods can be summarized as CNN can automatically learn features from large-scale data and generalize the results to unknown data of the same type. At present, CNN is the best way to extract image domain features, which greatly improves the accuracy of data classification. In this paper, CNN is used to classify.

Convolutional neural network includes input layer, hidden layer and output layer. The CNN structure is shown in Fig. 1.

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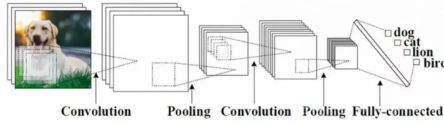


Figure 1: The CNN structures

Before the learning data is input into the convolutional neural network, the input data needs to be normalized in the channel or time/frequency dimension. If the input data is pixel, the original pixel values distributed in [0-255] can also be normalized to the range [0-1]. The standardization of input features is beneficial to improve the learning efficiency and performance of convolutional neural network.

Hidden layer includes convolution layer, pooling layer and full connection layer. The function of the convolutional layer is to extract features from the input data. It contains multiple convolutional kernels, and each element of the convolutional kernels corresponds to a weight coefficient and a bias vector, similar to neuron of a feed-forward neural network. After feature extraction in the convolution layer, the output feature graph will be transmitted to the pooling layer for feature selection and information filtering. The pooling layer contains a pre-configured pooling function that replaces the results of a single point in the feature graph with the feature graph statistics of its adjacent regions. The full connection layer is in the last part of the hidden layer of convolutional neural network and only transmits signals to other full connection layers.

For image classification problem, the output layer uses logic function or normalized exponential function (softmax function) to output classification label.

4 DATASET AND DATA PROCESSING

4.1 Dataset

In this project, we plan use CIFAR-10 database. CIFAR-10 is an established computer-vision dataset used for object recognition. It is a subset of the 80 million tiny images dataset and consists of 60,000 32x32 RGB color images containing one of 10 object classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck), with 6000 images per class. It was collected by Alex Krizhevsky, Vinod Nair, and Geoffrey Hinton. CIFAR-10 as shown in the Fig. 2.

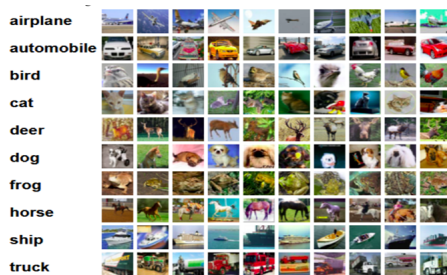


Figure 2: CIFAR - 10 examples

Compared with MNIST dataset, CIFAR-10 has the following differences:

- CIFAR-10 is a 3-channel color RGB image, while MNIST is a grayscale image.
- CIFAR-10 has a picture size of 32×32, while MNIST has a picture size of 28×28, slightly larger than MNIST.

Compared to handwritten characters, CIFAR-10 contains real objects in the real world, which are noisy and have different proportions and characteristics, making it difficult to identify. Direct linear models such as Softmax perform poorly on CIFAR-10.

4.2 Data processing

4.2.1 The image pixel normalization.

Cifar-10 is a 3-channel color picture with RGB values ranging from 0 to 255. For example, the RGB value of the tenth picture is [95,87,75]. This value is not conducive to training. In this paper, the RGB value is normalized by 255. For example, the RGB value of the tenth picture is normalized from [95,87,75] to [0.37254903, 0.34117648, 0.2941766]. The image normalization program and results are shown in the Fig. 3.

```
In [39]: Xtrain[10][10][10]
Out[39]: array([95, 87, 75], dtype=uint8)

In [40]: #Normalize the pixel value of the image
Xtrain_normalize = Xtrain.astype('float32')/255.0
Xtest_normalize = Xtest.astype('float32')/255.0
Xtrain_normalize[10][10][10]
Out[40]: array([0.37254903, 0.34117648, 0.2941766], dtype=float32)
```

Figure 3: The tenth picture pixel normalization program and results

4.2.2 The image's label normalization.

The CIFAR-10 dataset consists of 10 categories: "airplane", "Automobile", "bird", "Cat", "Deer", "dog", "frog", "horse", "ship", and "Truck". Use 0 to 9 to represent the 10 types. For example, the labels of the first 10 images in ciFAR-10 data set are [6, 9, 9, 4, 1, 1, 2, 7, 8, 3]. However, data labels from 0 to 9 are not easy to train.

One-hot encoding is a bit effective encoding, which mainly uses bit-state registers to encode each state. Each state has its own register bit, and only One bit is effective at any time. One-hot encoding uses 0 and 1 to represent some parameters and uses n-bit status register to encode N states. One-hot Encoding can handle discontinuous numeric characteristics. In our paper, one-hot Encoding is used to normalize the labels.

For example, the label value of the first picture is 6. Use one-hot Encoding normalized values for [0, 0, 0, 0, 0, 1, 0, 0, 0]. The label values and normalized results of the top 10 images are shown in Fig. 4.

```
In [12]: Ytrain[:10]
Out[12]: array([6, 9, 9, 4, 1, 1, 2, 7, 8, 3])

In [13]: Ytrain_onehot[:10]
Out[13]: array([[0., 0., 0., 0., 0., 1., 0., 0., 0.],
 [0., 0., 0., 0., 0., 0., 0., 0., 1.],
 [0., 0., 0., 0., 0., 0., 0., 0., 1.],
 [0., 0., 0., 1., 0., 0., 0., 0., 0.],
 [0., 1., 0., 0., 0., 0., 0., 0., 0.],
 [0., 1., 0., 0., 0., 0., 0., 0., 0.],
 [0., 0., 1., 0., 0., 0., 0., 0., 0.],
 [0., 0., 0., 0., 0., 0., 1., 0., 0.],
 [0., 0., 0., 0., 0., 0., 0., 1., 0.],
 [0., 0., 0., 1., 0., 0., 0., 0., 0.]])
```

Figure 4: The top 10 image's label normalization program and results

5 EVALUATION RESULTS

Our training and testing run on the CPU.

5.1 Data processing

In this paper, Accuracy and loss function are used to evaluate the model.

$$Accuracy = (TP + TN) / (P + N)$$

Where, TP is True positives, FP is False positives, FN is False negatives, and TN is True negatives. $P = TP + FP$, $N = FN + TN$.

5.2 Results

5.2.1 The original model.

After using two convolution layers and two pooling layers, the experiment entered the full connection layer with 128 neurons, and the output layer had a total of 10 units and 10 categories.

The value of Loss function after 150 Epochs is shown in Fig. 5. After 150 epochs, the training accuracy reached 0.61, as shown in Fig. 6.

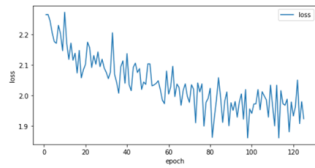


Figure 5: The value of Loss (150 epochs)

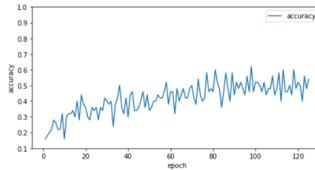


Figure 6: Training accuracy (150 epochs)

The accuracy of CIRAR-10 test is 0.60. The predictions for the top 10 graphs are shown in Fig. 7.



Figure 7: The predictions for the top 10 pictures (150 epochs)

As can be seen from Fig. 7, the prediction of the first, fourth and ninth pictures is wrong, while the prediction of the other pictures is correct.

With the increase of the number of epochs, the accuracy of training can reach 82% after 1220 epochs. After 1220 epochs, the accuracy and loss values are shown in Fig. 8 and Fig. 9.

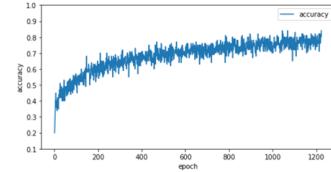


Figure 8: Training accuracy (1220 epochs)

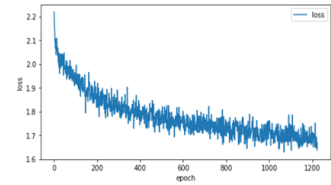


Figure 9: The value of Loss (1220 epochs)

when the model is used to test 10000 test data. The test accuracy is reached 79%. The prediction results of the first 10 pictures are shown in the Fig. 10. It can be seen from Fig. 10 that the predictions of other pictures are correct except the second one.



Figure 10: The predictions for the top 10 picture (1220 epochs)

5.2.2 1 full connection layer VS 2 full connection layers .

In this paper, the accuracy is improved by adding a full connection layer. One fully connected layer was used in the original model and two fully connected layers were used in the later experimental model. The accuracy of 1000 epochs is shown in Fig. 11.

After 1000 epochs training sessions. The accuracy of training can reach 91%. The accuracy of testing can reach 90.5%.

The prediction results of the first 10 graphs are shown in the Fig. 12. As can be seen from Fig. 12, top 10 picture predictions are all correct.

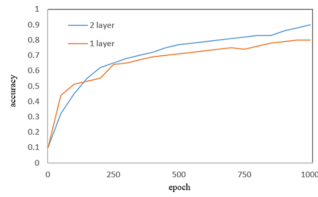


Figure 11: The accuracy of 1000 epoch (1 full connection layer VS 2 full connection layers)

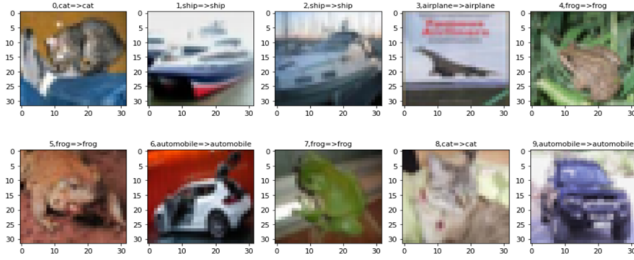


Figure 12: The predictions for the top 10 picture (1000 epochs and 2 full connection layers)

6 CONCLUSION

After training and testing CIFAR 10 with neural network, the test accuracy after 150 epochs reaches 0.60. As we can see from the

value of Loss function, its value has not reached convergence, so there is room for improvement of this neural network. Firstly, this paper increased the accuracy of training and testing by increasing the number of epochs. When the epochs were 1220, the accuracy of testing can reach 79%. Secondly, according to the experimental results, adding the full connection layer in CNN can also improve the accuracy of the test. When there are two fully connected layers, the test accuracy can reach 90.5% after 1000 epochs.

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